

Week 1 — Course Glossary

Pharma commercial vocabulary for data scientists

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Handout · Week 1 (Sep 14) · keep this open all semester

117 terms in nine groups. Bold terms are the ones you will be expected to use correctly from Week 2 onward.

How to use this glossary: read it once tonight (30 minutes), then return to it when a reading or a mentor uses a term you don't recognize. Definitions are written from the point of view of a commercial analytics team at a U.S. manufacturer; other countries differ, especially on pricing and access. Terms link to the week where they matter most.

1. Stakeholders and organizations

Term	Definition	Week
Manufacturer	The pharmaceutical or biotech company that develops, makes and markets a drug. In this course, the fictional Aspen Therapeutics (maker of GLUCOREN). "Commercial" = the sales, marketing, market-access, insights/analytics and commercial-operations functions.	Wk 1
Emerging biopharma (EBP)	Smaller companies (roughly <\$500M revenue or pre-revenue) that account for ~30% of launches but typically post smaller first-year sales; often partner with or sell to larger firms before commercialization.	Wk 1, 6
Payer	Whoever bears the cost of care: commercial insurers, self-insured employers, Medicare (federal, age 65+ and disabled), Medicaid (state/federal, low income), the VA and other government programs.	Wk 1, 2
Plan sponsor	The employer or government entity that funds a health plan; hires an insurer and/or PBM to administer it.	Wk 1
PBM (Pharmacy Benefit Manager)	Company that administers the pharmacy benefit for payers: builds formularies, adjudicates pharmacy claims in real time, negotiates rebates with manufacturers, contracts pharmacy networks, often owns mail-order and specialty pharmacies. Three PBMs handle ~80% of U.S. prescriptions.	Wk 1, 2, 6
HCP (Health-care professional / provider)	An individual clinician who can prescribe or administer: physician (MD/DO), nurse practitioner (NP), physician assistant (PA), pharmacist. Identified by NPI. The primary "customer" for promotion.	Wk 1, 4, 5
HCO (Health-care organization)	The organization an HCP works in: practice, clinic, hospital, health system. Has its own (Type 2) NPI.	Wk 1, 4
IDN (Integrated Delivery Network)	A health system that owns multiple sites of care (hospitals, clinics, sometimes a health plan and pharmacies) and increasingly makes formulary and protocol decisions centrally (e.g., Kaiser, Intermountain, HCA).	Wk 1, 4
KOL (Key Opinion Leader)	A physician-researcher whose publications, guideline work and speaking influence peers' prescribing. Engaged by medical affairs, not sales.	Wk 1, 7
MSL (Medical Science Liaison)	Field-based medical-affairs professional who exchanges scientific information with HCPs and KOLs. Non-promotional by law; their field notes are a rich NLP source.	Wk 1, 7
Wholesaler / distributor	Buys drugs from manufacturers and sells to pharmacies and hospitals. McKesson, Cencora (formerly AmerisourceBergen) and Cardinal Health move >90% of U.S. volume.	Wk 1
GPO (Group Purchasing Organization)	Negotiates purchasing contracts on behalf of member hospitals and clinics (Vizient, Premier, HealthTrust). Relevant to hospital-administered drugs.	Wk 1
Specialty pharmacy	Pharmacy that dispenses high-cost, complex therapies (biologics, oncology, rare disease) and provides adherence support; often PBM-owned. A narrow network can restrict where a drug is dispensed.	Wk 1, 2

Hub / patient services	Manufacturer-funded program that helps patients with benefits verification, prior-authorization support, copay assistance and adherence nursing. Generates patient-level data outside claims.	Wk 1, 5
Contract pharmacy (340B)	A retail or mail pharmacy that dispenses 340B-priced drugs on behalf of a covered entity (hospital/clinic).	Wk 1
Policymaker / regulator	FDA (approval, labeling, promotion rules), CMS (Medicare/Medicaid, IRA price negotiation), HRSA (340B), FTC (competition), state legislatures (PBM regulation).	Wk 1, 6

2. Products, lifecycle and regulation

Term	Definition	Week
Brand vs. generic vs. biosimilar	Brand: the originator product sold under a trademark. Generic: chemically identical small molecule sold after patent expiry, typically at a 80–90% discount. Biosimilar: highly similar version of a biologic; erosion is slower and less complete than for generics.	Wk 1, 6
Small molecule vs. biologic	Small molecules are chemically synthesized pills/capsules (cheap to make, easy to copy). Biologics are large molecules made in living cells (injectables/infusions, higher COGS, harder to copy).	Wk 1
Indication	The disease or condition a drug is approved to treat, as stated in the FDA label. Promotion is limited to approved indications (“on-label”).	Wk 1, 2
Label (prescribing information)	The FDA-approved document describing indications, dosing, safety and clinical data. The legal boundary of promotional claims (see MLR).	Wk 1, 7
Orphan drug	A drug for a rare disease (U.S.: <200,000 patients). Orphan designation brings exclusivity and incentives; orphan drugs were 53% of U.S. launches in the past five years.	Wk 1
Lifecycle: pre-launch / launch / in-line / LOE	The commercial phases of a brand. Each phase asks different analytics questions and uses different data (see Week 1 slides).	Wk 1, 6
LOE (Loss of Exclusivity)	The date patents and regulatory exclusivities expire and generics/biosimilars can enter. Small-molecule brands typically lose 80–90% of volume within a year.	Wk 1, 6
Launch archetype	IQVIA's framework classifying launches by unmet need × product differentiation: Science Sells, Market Shaping, Emphasize the Difference, Who Benefits.	Wk 1, 6
Launch curve / analog	The trajectory of a new brand's prescriptions over its first 1–3 years; forecast by reference to historical “analog” launches with similar characteristics.	Wk 6
Line of therapy (LOT)	The sequence position of a treatment regimen (1L, 2L, 3L+) in a patient's journey; derived from claims using an algorithm.	Wk 2
Standard of care (SoC)	The treatment most physicians would use for a given patient; the comparator a new brand has to displace.	Wk 1, 2
Clinical trial phases	Phase 1 (safety, small n), Phase 2 (efficacy signal), Phase 3 (pivotal, large), Phase 4 (post-marketing). ~11% composite success from Phase 1 to submission; ~\$3.1B average cost per approved therapy.	Wk 1
Sunshine Act / Open Payments	Federal law requiring manufacturers to report payments and transfers of value to physicians and teaching hospitals; public dataset.	Wk 1, 4
IRA / Medicare price negotiation / MFP	The Inflation Reduction Act (2022) allows CMS to negotiate a “maximum fair price” for selected high-spend Medicare drugs, effective 2026 for the first 10.	Wk 1, 6

3. The prescription and its metrics

Term	Definition	Week
Rx	Shorthand for a prescription.	Wk 1
NRx (New prescriptions)	Prescriptions that are not refills. Note: a patient switching back to a brand generates an NRx; “new” ≠ “new to brand.”	Wk 1, 3
TRx (Total prescriptions)	NRx + refills in a period. The standard volume measure and the response variable in most promotion-response models.	Wk 1, 3
NBRx (New-to-brand prescriptions)	Prescriptions for patients with no claim for the brand in a look-back window (typically 12 months). Requires longitudinal patient data; the cleanest launch	Wk 1, 2, 6

	metric.	
Normalized / extended-unit TRx	TRx adjusted for days' supply (e.g., a 90-day script = 3 × 30-day equivalents) so mail-order and retail are comparable.	Wk 2, 3
Share (TRx / NRx / NBRx share)	Brand ÷ market. Always ask how the market is defined (molecules, indication, line of therapy, channel).	Wk 1, 2
Source of business	Where a brand's new patients come from: naïve (untreated), switches from competitor A/B, add-ons. From longitudinal claims.	Wk 2
Persistence	How long a patient stays on therapy before a gap exceeding a threshold (e.g., 60 days). Reported as median days or % persistent at 6/12 months.	Wk 2, 6
Adherence • PDC • MPR	Whether patients take the drug as prescribed. Proportion of days covered (PDC) = days with drug on hand ÷ days in period; medication possession ratio (MPR) is the older, refill-based version. ≥80% PDC is the common “adherent” threshold.	Wk 2, 5
Abandonment	A prescription that was adjudicated but never picked up (the claim is reversed). Strongly driven by out-of-pocket cost.	Wk 1, 2
Writer	An HCP who wrote at least one prescription for a product/market in a period. “Writer counts” are a key launch metric.	Wk 4
Decile	HCPs ranked by prescribing volume and split into ten groups; decile 10 = highest volume. The traditional basis for targeting.	Wk 4, 5
Reach and frequency	Reach: % of target HCPs contacted at least once in a period. Frequency: contacts per reached HCP. The two dials of field-force deployment.	Wk 3, 5
Call / detail / e-detail	A sales-rep visit to an HCP (call); the product discussion within it (detail); a remote/digital version (e-detail).	Wk 3, 5
Samples	Free product given to HCPs for patients; a promotional channel with its own regulation (PDMA).	Wk 3
DTC (Direct-to-consumer)	Advertising to patients (TV, digital). Permitted in the U.S. and New Zealand only; heavily regulated (fair balance).	Wk 3

4. Pricing, contracting and access

Term	Definition	Week
WAC (Wholesale Acquisition Cost)	The manufacturer's published list price to wholesalers, before discounts. The starting point of every gross-to-net calculation.	Wk 1
AWP (Average Wholesale Price)	A legacy benchmark (~WAC × 1.2) still used in some pharmacy reimbursement formulas.	Wk 1
ASP (Average Sales Price)	Manufacturer's average net price to all purchasers; Medicare Part B reimburses physician-administered drugs at ASP + 6%.	Wk 1
Gross-to-net (GTN)	The sum of all deductions from gross (WAC) sales to net revenue: rebates, chargebacks, fees, statutory discounts, copay assistance, returns. Expressed as % of gross. Ranges from ~15% to >70% by brand.	Wk 1, 3
Net price	Revenue per unit actually retained by the manufacturer after GTN deductions. Confidential; estimated by analysts from public filings.	Wk 1
Rebate	A payment from the manufacturer to a PBM/payer, made in arrears (typically quarterly), in exchange for formulary placement or market-share performance.	Wk 1
Chargeback	Payment from manufacturer to wholesaler covering the difference between WAC and a lower contract price the wholesaler honored for a customer (GPO, 340B, VA).	Wk 1
Medicaid Drug Rebate Program	Statutory rebate manufacturers must pay on Medicaid utilization: the greater of 23.1% of AMP or best price, plus an inflation penalty.	Wk 1
340B program	Federal program requiring manufacturers to sell outpatient drugs at deeply discounted “ceiling prices” to qualifying safety-net hospitals and clinics (“covered entities”). ~\$180–200B at list in 2025.	Wk 1
Copay card / copay assistance	Manufacturer program that pays part of the patient's out-of-pocket cost for a branded drug (commercial patients only; prohibited in Medicare).	Wk 1, 5
Accumulator / maximizer	Payer programs that stop copay-card payments from counting toward the patient's deductible (accumulator) or set the copay to the card's maximum (maximizer).	Wk 1

Formulary	The payer/PBM list of covered drugs, organized in tiers. Being “on formulary” at a preferred tier is the central market-access objective.	Wk 1, 2
Tier	Formulary level determining patient cost sharing: Tier 1 generics (lowest copay), Tier 2 preferred brands, Tier 3 non-preferred brands, Tier 4/5 specialty (often coinsurance).	Wk 1, 2
Utilization management (UM)	Payer tools that restrict use: prior authorization (PA — physician must justify), step therapy / step edit (must fail a cheaper drug first), quantity limits.	Wk 1, 2
Covered lives	Number of people enrolled in plans where a drug has a given formulary status; the unit of market-access reporting.	Wk 1, 6
Buy-and-bill	The model for physician-administered drugs: the practice buys the drug (via wholesaler/GPO), administers it, and bills the medical benefit (HCPCS J-code). Contrast with the pharmacy benefit.	Wk 1, 2
Medical benefit vs. pharmacy benefit	Two different insurance “buckets.” Infused/injected drugs administered by HCPs run through the medical benefit (medical claims); self-administered drugs through the pharmacy benefit (pharmacy claims).	Wk 1, 2
HTA (Health Technology Assessment)	Formal evaluation of a drug’s clinical and economic value used by payers (mainly ex-U.S.: NICE in the UK, G-BA in Germany; ICER in the U.S.).	Wk 1
HEOR	Health economics and outcomes research: the function that builds cost-effectiveness models and value dossiers for payers.	Wk 1, 2

5. Codes and identifiers

Term	Definition	Week
ICD-10-CM	International Classification of Diseases, 10th revision, clinical modification: ~70,000 diagnosis codes on U.S. medical claims (e.g., E11.9 type 2 diabetes without complications; C83.3 diffuse large B-cell lymphoma).	Wk 1, 2
CPT	Current Procedural Terminology (AMA): 5-digit codes for services and procedures (99213 office visit; 96413 chemotherapy infusion).	Wk 2
HCPCS Level II	Healthcare Common Procedure Coding System: alphanumeric codes for products and supplies on medical claims. J-codes identify physician-administered drugs (J9310 rituximab); Q- and S-codes are also common.	Wk 2
NDC (National Drug Code)	10- or 11-digit code (labeler-product-package) identifying a specific drug, strength, form and package size. On every pharmacy claim. Maintained by FDA.	Wk 1, 2
NPI (National Provider Identifier)	10-digit unique identifier for HCPs (Type 1) and organizations (Type 2), issued by CMS; public registry (NPPES) includes specialty and address.	Wk 1, 2, 4
DEA number	Registration number required to prescribe controlled substances; sometimes used as an alternate HCP key.	Wk 2
Taxonomy code	The specialty classification attached to an NPI (e.g., 207RE0101X endocrinology).	Wk 4
DRG	Diagnosis-related group: the bundle Medicare uses to pay hospitals for inpatient stays.	Wk 2
LOINC • SNOMED • RxNorm	Standard vocabularies for lab tests (LOINC), clinical concepts (SNOMED CT) and drug names (RxNorm), mostly in EHR data.	Wk 2
Place of service (POS)	Two-digit code on medical claims for the setting (11 office, 22 hospital outpatient, 23 ER).	Wk 2

6. Data sources

Term	Definition	Week
Claims data (medical / pharmacy)	Billing records submitted to payers. Medical claims carry diagnoses and procedures; pharmacy claims carry dispensed drugs (NDC), days' supply, and cost splits.	Wk 2
Open vs. closed claims	Closed: all claims for enrolled members of specific payers (complete but limited coverage). Open: claims collected from clearinghouses and pharmacies across many payers (broad but incomplete patient capture).	Wk 2
Longitudinal patient data	De-identified patient-level data linking claims over time via a token, enabling	Wk 2

(LPD / APLD)	journeys, NBRx, persistence and source of business.	
Rx panel / retail audit	Prescription data collected from pharmacies (~90%+ of U.S. retail) and projected to 100%; the source of NRx/TRx at the HCP level (e.g., IQVIA Xponent-type data).	Wk 2, 3
DDD (Drug Distribution Data)	Sales-to-outlet data from wholesalers, showing units shipped to pharmacies, hospitals and clinics; the basis for sales-force compensation in many companies.	Wk 2
EHR / EMR	Electronic health/medical records: clinical detail (labs, vitals, notes, staging) not present in claims; harder to license and messier.	Wk 2, 7
Syndicated data	Standardized datasets licensed by many companies from vendors (IQVIA, Komodo, Veeva Compass, Symphony) as opposed to custom studies.	Wk 2
Projection	Statistical scaling of a sample (e.g., 90% of pharmacies) to a national estimate; introduces error that varies by geography and product.	Wk 2
RWD / RWE	Real-world data (claims, EHR, registries) and the real-world evidence derived from it, increasingly used in regulatory and payer decisions.	Wk 2
De-identification / tokenization / HIPAA	Removing identifiers and replacing patient identity with a consistent token so records can be linked without exposing who the patient is. HIPAA governs protected health information (PHI).	Wk 2
CRM (e.g., Veeva)	The system where reps log calls, samples and HCP details; the source of promotional activity data.	Wk 3, 5
Master data (HCP / HCO / product)	Reference tables that reconcile identifiers across sources (NPI ↔ CRM ID; NDC ↔ molecule ↔ brand). Poor master data is the #1 cause of broken analyses.	Wk 2

7. Commercial functions and promotion

Term	Definition	Week
Brand team	Cross-functional team (brand lead, marketing, medical, access, analytics, finance) accountable for a product's P&L.	Wk 1
Field force / sales force	Reps organized by geography (territories → districts → regions) and often by specialty; sized and aligned using analytics (SFE).	Wk 3, 5
SFE (Sales-force effectiveness)	Analytics on territory alignment, targeting, call planning, incentive compensation.	Wk 4, 5
Targeting / segmentation	Choosing which HCPs to promote to (targeting) and grouping them by behavior, potential and attitudes (segmentation).	Wk 4
Omnichannel	Coordinating personal (rep) and non-personal (email, web, ads, conferences) channels around each HCP rather than running them in silos.	Wk 5
NBA (Next Best Action)	A decision system that recommends the next channel, message and timing for each HCP, balancing response likelihood against fatigue and constraints.	Wk 5
MMM (Marketing / Media Mix Modeling)	Regression-type models of sales as a function of channel spend with carryover (adstock) and saturation, used to measure ROI and reallocate budget.	Wk 3
Promotional intensity	Total promotional spend ÷ net sales; benchmark for whether a brand is over- or under-invested.	Wk 1, 3
Market access	The function that secures formulary coverage and reimbursement; negotiates contracts with payers, PBMs and GPOs.	Wk 1, 6
Medical affairs	Non-promotional scientific function: MSLs, publications, medical information, HEOR, investigator-initiated studies.	Wk 1, 7
MLR (Medical-Legal-Regulatory) review	Mandatory review of every promotional asset for label consistency, fair balance and legal risk; a bottleneck that GenAI tools increasingly assist.	Wk 7
Advisory board (ad board)	A paid panel of HCPs or patients convened to give a company advice; “synthetic ad boards” use LLM personas to pre-test questions.	Wk 7
Compliance	Internal function ensuring promotion, payments to HCPs and data use follow FDA, OIG, anti-kickback and privacy rules.	Wk 1
Pharmacovigilance (PV)	Monitoring and reporting of adverse events; any AE mentioned to a rep or on social media must be reported within regulatory timelines.	Wk 7

8. Analytics and AI terms used in this course

Term	Definition	Week
Adstock / carryover	The lagged, decaying effect of promotion on sales in MMM.	Wk 3
Saturation / Hill curve	Diminishing returns to promotional spend, modeled with a Hill or logistic function.	Wk 3
Uplift / incrementality	The causal effect of a promotion on an HCP or patient relative to not promoting; measured by experiments or uplift models.	Wk 3, 5
Patient-finding	Predictive models that identify patients likely to have an undiagnosed condition or to be eligible for a therapy, from claims patterns.	Wk 2, 5
Epidemiology-based forecast (epi funnel)	Forecast built from population → prevalence → diagnosed → treated → eligible → brand share → compliance → units.	Wk 6
Bass diffusion	A classic model of new-product adoption (innovators + imitators) used for launch curves.	Wk 6
Foundation model (time series / language)	Large pre-trained models (TimeGPT, Chronos; BERT/GPT families) applied zero-shot or fine-tuned to pharma data.	Wk 6, 7
Embeddings / RAG	Vector representations of text and retrieval-augmented generation — the building blocks of LLM systems over documents (labels, field notes, formulary PDFs).	Wk 7
Agentic AI	AI systems “empowered to perform dedicated tasks autonomously” (IQVIA): an orchestrator routes a request to specialist agents that use tools and data, then synthesizes the answer. Human review remains in the loop for decisions.	Wk 1, 10+
Healthcare-grade AI	IQVIA's term for AI engineered to the precision, traceability and privacy standards required in life sciences.	Wk 1
Human in the loop	A design principle in which a person reviews AI output before it is used for a decision or a business process — the norm in regulated commercial functions.	Wk 7
Data readiness / sufficiency	Whether an organization's data is structured, connected and accessible enough to support AI at scale; only 11% of surveyed commercial leaders call theirs “fully sufficient.”	Wk 1, 2

9. Fictional entities used in this course

Term	Definition	Week
Aspen Therapeutics	The fictional mid-size manufacturer whose brand teams we work for throughout the course.	All
GLUCOREN	Fictional branded once-daily oral agent for type 2 diabetes, three years post-launch, ~\$650M net sales; the running example for stakeholder, P&L and access discussions.	Wk 1+
Brand X	Fictional oral chronic-indication brand used in the Week 3 MMM homework.	Wk 3
DLBCL case	Synthetic diffuse large B-cell lymphoma claims dataset used in Week 2 for cohorting and line-of-therapy analysis.	Wk 2

Ten terms to test yourself on before Week 2

Cover the definitions and explain each of these to a classmate in one sentence: PBM • gross-to-net • NBRx • formulary tier • prior authorization • NDC • NPI • ICD-10 • LOE • buy-and-bill.